

Sumit Srivastava

AI & Engineering Leader | PhD, Human-AI Dialogue

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Nationality: Dutch

Summary

AI and engineering leader with 8+ years building and leading large-scale production systems, and a PhD in human-AI dialogue (defence 2026). Combines hands-on LLM and NLP work, including RAG, controllable text generation, and evaluation design, with a track record of leading engineering teams and shipping high-throughput systems at 1.5M requests/second and sub-2ms p99 latency.

Skills

AI / ML: LLMs, RAG, controllable text generation, evaluation (LLM-as-a-judge, red teaming), dialogue systems, NLP, PyTorch, spaCy, Rasa

Languages: Python, Java, TypeScript / Node.js

Data & Infrastructure: MySQL, PostgreSQL, MongoDB, Cassandra, AWS, Azure, Redshift, Git, Linux

Leadership: Technical strategy, architecture, team leadership, mentoring, hiring

Experience

Senior Backend Engineer

Tibo Energy

Feb 2025 – Present

The Netherlands

- Design, build, and maintain features for an industrial energy-management platform.
- **Stack:** TypeScript, Node.js, MySQL, Cassandra.

Doctoral Researcher, Applied AI & NLP

University of Twente / NL4XAI (Marie Skłodowska-Curie)

Sep 2020 – Present

The Netherlands

- Designed conversational agents that adapt language to individual users; built controllable LLM generation methods for lexical alignment (under review, SigDial 2026).
- Ran controlled user studies measuring comprehension, trust, persuasion, cognitive load, and information uptake in human-AI dialogue.
- Supervised eight MSc/BSc students; published at ACM IUI, ACM UMAP, and *Computer Speech & Language*.

Team Manager & Technical Lead

MiQ Digital

Jan 2017 – Sep 2019

India

- Built and mentored a team of seven engineers across seven production services.
- Re-architected a real-time ad-bidding platform to 1.5M requests/second at sub-2ms p99 latency via event-driven design and database optimisation.
- Defined company-wide code and documentation standards; contributed performance fixes to an open-source PostgreSQL library.

Senior Software Engineer & Software Engineer

MiQ Digital

Sep 2013 – Dec 2016

India

- Built a real-time bid-modification service handling up to 1.5M requests/second.
- Designed a multi-protocol data-ingestion service (12+ protocols, 30+ sources) shipping up to 90 TB/day to AWS S3 and Redshift.

Earlier: Software Engineer, Apostek Softwares (2012–2013).

Advisory & Leadership Roles (concurrent)

CTO (part-time)

BrainBite

Dec 2024 – Sep 2025
The Netherlands (Remote)

- Set technical direction and owned architecture for the company's AI vertical.
- Led reliability improvements across LLM-based content-generation modules.
- Established automatic and manual evaluation methodologies, including LLM-as-a-judge and red-teaming, for the AI systems.

CTO (freelance, intermittent)

ad:personam GmbH

2021 – Present
Remote

- Built the web application (starting from the MVP) for a self-serve programmatic-advertising DSP, including third-party integrations (Xandr, Stripe, OpenAI).
- Owned CI/CD, AWS/Azure infrastructure, security, privacy, and cost policies; guided technical hiring.

Selected Publications

- **Weighted Decoding for Controlled Lexical Alignment in LLM-based Dialogues.** *SIGDIAL*, 2026 (under review).
- Srivastava, S., Wentzel, S., Theune, M., & Catala, A. (2024). Measuring and Implementing Lexical Alignment: A Systematic Literature Review. *Computer Speech & Language*, Elsevier.
- Srivastava, S., Theune, M., & Catala, A. (2023). The Role of Lexical Alignment in Human Understanding of Explanations by Conversational Agents. *ACM IUI*, 423–435.
- Srivastava, S., Theune, M., Catala, A., & Reed, C. (2024). Trust in a human-computer collaborative task with or without lexical alignment. *ACM UMAP*

Education

Ph.D., Computer Science (Human–Computer Interaction)

University of Twente

2020 – 2026
The Netherlands

M.Sc., Artificial Intelligence (*Cum Laude*)

KU Leuven

2019 – 2020
Belgium

M.Tech., Information Technology (CGPA 9.2/10)

IIT Allahabad

2010 – 2012
India

B.Tech., Computer Science

Uttar Pradesh Technical University

2006 – 2010
India